

# The Accessibility Paradox: When LLM Simplification for Students with Learning Difficulties Amplifies Strategic Misconceptions in Computer Science Education

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## Abstract

Large language models (LLMs) are increasingly used as assistive technology by students with reading and attention difficulties (e.g., dyslexia, ADHD), who rely on them as patient, private tutors capable of simplifying complex texts. However, it remains an open question whether this simplification compromises conceptual accuracy. This paper empirically investigates the accessibility paradox: the hypothesis that LLMs' attempts to be inclusive and simple result in an increased rate of strategic misconceptions, i.e., subtle, conceptually flawed, yet persuasively formulated responses that are particularly dangerous for vulnerable learners who already expend significant cognitive effort on text decoding and plausibly have reduced spare capacity for parallel fact-verification. We introduce AccessTrap-CS, a benchmark corpus of 630 responses generated by seven open-weight LLMs across 30 core computer science concepts spanning five subfields (6 per subfield). Each concept was prompted under three conditions: (a) a standard explanation, (b) a simplified explanation simulating interaction with a student with reading difficulties, and (c) a highly simplified explanation simulating interaction with a student with severe attention difficulties. Responses were evaluated for strategic misconception rate (SMR), dangerous analogy rate (DAR), and oversimplification error rate (OER), a novel error category capturing inaccuracies that arise exclusively from the simplification process. Statistical comparisons across models and conditions employ one-way ANOVA with Tukey HSD post-hoc correction and chi-square tests. Based on our findings, we propose three accessibility-first design principles for AI tutoring interfaces that visually signal model uncertainty without relying on text-based disclaimers, thereby protecting the very learners these systems aim to serve.

**Keywords:** large language models; inclusive education; strategic misconceptions; accessibility; dyslexia; ADHD; assistive technology; AI safety; computer science education; oversimplification errors

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## 1. Introduction

Large language models (LLMs) are rapidly being adopted as assistive technology by students with reading and attention difficulties [1–3]. For learners with dyslexia, who struggle with text decoding, and students with ADHD, who face sustained attention challenges, LLMs offer a compelling value proposition: a patient, private tutor, available on demand, capable of simplifying complex academic texts in infinitely flexible ways [4,5].

Unlike human tutors, LLMs are available around the clock, never judge repeated questions, and can rephrase explanations until something clicks. It is therefore unsurprising that general-purpose assistants such as ChatGPT and Claude are increasingly used in this capacity, often without institutional oversight or teacher involvement. The same pattern is visible in computing education at large, where generative AI has moved from a curiosity to a routine study aid within a few years [6–8]. Early evidence on AI tutoring is encouraging [9,10], but it also carries a warning: unrestricted access to a generative assistant can raise performance while the assistant is available and lower it once the assistance is removed [11]. What the assistant says, and how it says it, matters for what the student actually learns.

However, this promising picture conceals a potentially dangerous blind spot. The very features that make LLMs attractive for students with learning difficulties, such as simplified language, short sentences, concrete analogies, and the avoidance of technical jargon, may inadvertently introduce subtle but pedagogically significant errors. We term this phenomenon the accessibility paradox: the hypothesis that LLMs' well-intentioned attempts to be inclusive and accessible result in an increased rate of what we call strategic misconceptions: subtly flawed yet persuasively formulated responses that are particularly dangerous for vulnerable learners.

We define a strategic misconception as an explanation that is conceptually coherent, internally consistent, and often intuitively appealing, but that nonetheless contains a fundamental misunderstanding of the target concept. Unlike random errors or obvious falsehoods, strategic misconceptions are dangerous precisely because of their plausibility: a student who encounters one is not alerted to an error but is instead led to construct a mental model that is systematically wrong. This definition draws on the conceptual-change tradition in science and mathematics education [12–14] and on the misconception literature in computing [15,16], which consistently shows that intuitive but incorrect prior conceptions are among the hardest to dislodge through subsequent instruction. Two further findings make plausible-but-wrong explanations especially hazardous: fluent, easy-to-process statements are judged more likely to be true [17,18], and people routinely overestimate how well they understand mechanisms after reading a satisfying but shallow explanation [19]. For students with dyslexia and ADHD, who already expend significant cognitive effort on text decoding and attention regulation [20–22], the spare capacity for parallel fact-verification is plausibly reduced. We stress that this is a situational vulnerability arising from processing load and missing scaffolding, not a deficit in intelligence, and that it applies in some measure to any novice reading under high load.

The computer science (CS) education context makes this problem particularly acute. CS concepts are often abstract, hierarchical, and resistant to faithful simplification through everyday analogy [16,23,24]. A concept such as a semaphore, a deadlock, or virtual memory cannot be accurately captured by a single intuitive metaphor without loss of essential technical content. Yet, under pressure to simplify, LLMs frequently resort to analogies that are intuitively appealing but technically misleading. For example, comparing a semaphore to a traffic light fails to convey atomicity and counting semantics, and describing a deadlock as two people stuck in a doorway omits resource holding and circular wait conditions. These errors are not isolated or accidental; they are systematic products of the simplification process itself. We term this subclass oversimplification errors (OER): errors that would not appear in a standard explanation but emerge exclusively as a consequence of accessibility-driven simplification.

This trade-off between warmth and accuracy is not unique to accessibility contexts. Ibrahim et al. [25] demonstrated that fine-tuning LLMs to produce warmer, more empathetic responses consistently increases error rates across model architectures, with warm models showing substantially higher rates of factual inaccuracies and sycophantic valida-

tion of incorrect user beliefs, particularly when users expressed emotional vulnerability. This fits a broader line of evidence that preference-based alignment can reward agreeable, confident-sounding answers over correct ones [26–28]. Accessibility-driven simplification represents an analogous pressure: models instructed to be maximally clear and engaging may sacrifice technical precision in ways that standard evaluation benchmarks fail to detect [29].

Prior work has documented the benefits of LLMs for students with disabilities [1,30], has studied factual errors introduced by automatic text simplification in general-purpose text [31,32], and has separately studied misconceptions in CS education [15,23]. However, to the best of our knowledge, no study has empirically investigated whether accessibility-driven simplification systematically increases the rate of strategic misconceptions in LLM-generated CS explanations. This gap is critical: if simplification amplifies errors, then current assistive technology practices may be inadvertently harming the very population they aim to serve.

### *Research questions and hypotheses*

The study is organised around four research questions, each paired with a directional hypothesis.

- RQ1. Does accessibility-driven simplification increase the strategic misconception rate? H1:  $SMR(B) > SMR(A)$  and  $SMR(C) > SMR(A)$ .
- RQ2. Does it increase the use of dangerous analogies? H2:  $DAR(C) \geq DAR(B) > DAR(A)$ .
- RQ3. Does simplification create errors that are absent from the standard explanation of the same concept, and do these grow with simplification intensity? H3:  $OER(B) > 0$ ,  $OER(C) > 0$ , and  $OER(C) > OER(B)$ .
- RQ4. Are these effects moderated by model and CS subfield? This question is exploratory; we make no directional prediction.

In this paper, we present AccessTrap-CS, a benchmark corpus of 630 responses generated by seven open-weight LLMs across 30 core computer science concepts spanning five subfields: algorithms and data structures, programming languages and paradigms, databases, computer networks, and operating systems (6 per subfield). Each concept was prompted under three conditions: (a) a standard explanation suitable for a first-year CS student (baseline), (b) a simplified explanation simulating interaction with a student with reading difficulties (dyslexia condition), and (c) a highly simplified explanation simulating interaction with a student with severe attention difficulties (ADHD condition). Responses were evaluated using an LLM-as-judge approach [33], with Qwen3:235b serving as the expert evaluator.

Our contributions are as follows. First, we provide empirical evidence on the accessibility paradox, testing whether accessibility-driven simplification, which increases readability and analogy frequency, also systematically introduces conceptual errors, including errors that do not occur under standard prompting conditions. Second, we introduce a novel error taxonomy: oversimplification errors (OER) as a distinct error category that emerges exclusively as a product of the simplification process and would not appear in technically precise explanations. Third, we present the first systematic analysis of dangerous analogies in LLM-generated CS content under accessibility prompts, documenting cases in which commonly generated analogies misrepresent critical technical properties of CS concepts. Fourth, we propose three accessibility-first design principles for AI tutoring interfaces that visually signal model uncertainty and analogy status without relying on text-based disclaimers, thereby protecting the very learners these systems aim to serve.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents quantitative and qualitative results. Section 5 discusses the implications of our findings. Section 6 proposes three design principles. Section 7 acknowledges limitations. Section 8 concludes.

## 2. Background and Related Work

The accessibility paradox sits at the intersection of several research streams that have developed largely independently: (1) LLMs as assistive technology for students with learning difficulties, (2) the reliability of LLMs and the pressures that trade accuracy for user-facing qualities, (3) text simplification and its pedagogical trade-offs, (4) the cognitive mechanisms that make fluent explanations persuasive and hard to verify, (5) analogies and misconceptions in computer science education, and (6) the methodology of LLM-based evaluation. This section reviews each stream and states precisely what it does and does not tell us about the accessibility paradox.

### 2.1. LLMs as Assistive Technology for Dyslexia and ADHD

Dyslexia is a neurodevelopmental difference characterised by difficulties with accurate and fluent word recognition and spelling, typically associated with weaknesses in the phonological component of language [20,34]. It persists into adulthood and into higher education, where reading loads are heavy and time-constrained. ADHD is characterised by persistent inattention and/or hyperactivity-impulsivity [21,35], and meta-analyses report working-memory impairments in affected children [22]; comparable working-memory weaknesses are reported for children with learning difficulties more broadly [36]. These are group-level tendencies. Individual profiles vary widely, and the two students in our prompts are operationalisations of an interaction style (simplified language, brevity), not clinical simulations. We return to this point in Section 7.

For learners with dyslexia, LLMs can rephrase dense academic prose into shorter sentences, replace rare vocabulary with common alternatives, and maintain semantic coherence without the social cost of asking a human tutor for repeated clarification [1]. Earlier work on text presentation and simplification for dyslexia showed that both vocabulary choices and typography measurably affect readability and understandability [4,37]. For students with ADHD, whose attention and working-memory limitations make extended technical reading challenging [21], LLMs offer on-demand summarisation and task segmentation [2]. Empirical evidence for the ADHD case specifically remains thin and largely anecdotal or self-reported; disability-centred studies of LLMs emphasise both real benefits and concerns about reliability and representation [30].

Broad overviews frame LLMs as an opportunity for personalised support alongside risks of inaccuracy and over-reliance [3]. In computing education, LLMs solve introductory programming tasks [6], and working groups document how quickly these tools have entered practice and how much of the pedagogical evidence is still missing [7,8]. Controlled studies suggest that LLM-generated help can match human-authored help on mathematics practice [9] and that a carefully designed AI tutor can outperform in-class active learning in a physics course [10]. Crucially, a field experiment in high-school mathematics found that free-form generative assistance improved performance during practice but harmed later unassisted performance, whereas a version with pedagogical safeguards largely mitigated the harm [11]. The design of what the assistant produces is therefore a first-order determinant of learning, not a cosmetic choice.

Studies on assistive text simplification have evaluated outcomes mainly through readability measures and user satisfaction [4,5], not through the conceptual accuracy of simplified technical content. No study to date has examined whether LLM-generated

simplifications for learners with reading or attention difficulties introduce novel errors that are absent in standard explanations. The gap is non-trivial: the mechanisms that make simplification beneficial (reduced decoding effort, shorter sentences, concrete analogies) may simultaneously suppress the nuanced qualifications and precise technical language that prevent conceptual misunderstanding.

## 2.2. LLM Reliability, Sycophancy, and the Warmth-Accuracy Trade-Off

LLMs produce fluent text that is not necessarily faithful to fact or to a source [38,39]. Two properties of this literature matter here. First, errors are not confined to obscure facts: models can reproduce popular misconceptions, and on the TruthfulQA benchmark the largest models were often among the least truthful because they imitate widespread human falsehoods more faithfully [29]. This is directly relevant to analogies, which are precisely the popular, widely repeated framings of technical ideas. Second, factual quality is prompt-dependent: the same model can be accurate under one instruction and misleading under another.

Instruction-following and preference-based training optimise for outputs that human raters approve [26]. Approval and correctness are correlated but not identical. Models trained this way exhibit sycophancy, tailoring answers to the beliefs or expectations of the user, and preference data itself can reward it [27,28].

Ibrahim et al. [25] demonstrated that fine-tuning models to produce warmer, more empathetic responses increases error rates across architectural families, with warm models showing a statistically significant tendency toward sycophantic validation of incorrect user beliefs. This trade-off is not an implementation flaw but a plausible consequence of alignment pressure: models optimised for human preference judgments learn that users respond positively to agreeable, confident-sounding explanations, even when those explanations are technically incomplete.

Where warmth optimisation prioritises affective alignment, simplification prioritises cognitive accessibility. The two are not identical: a simplified explanation need not be warm, and a warm explanation need not be simple. However, they share a common mechanism: both pressures push models away from precise, qualified, context-dependent technical statements and toward concrete, declarative, analogy-laden statements that are easier to process but more likely to be wrong in subtle ways.

We propose the term oversimplification error (OER) to capture a specific subclass of simplification-induced inaccuracies: errors that would not appear in a standard, technically precise explanation of the same concept but emerge exclusively as a product of the simplification process. OERs are distinct from random factual errors, such as misstating the year of an algorithm's invention, in that they arise systematically from the interaction between the concept's inherent complexity and the pressure to reduce that complexity. A standard explanation of semaphores, for example, would specify atomicity, counting semantics, and the distinction between blocking and busy-waiting. A simplified explanation that omits atomicity is not merely incomplete; it actively misleads by suggesting that semaphores are just integer variables with special names.

Widely used evaluations measure knowledge or truthfulness under a fixed, neutral prompt [29]. A model can therefore score well on standard tests and still degrade when told to speak to a particular kind of learner. AccessTrap-CS is designed to expose exactly this audience-conditioned degradation.

A natural response is to have models flag their own uncertainty. Verbalised model confidence is often overconfident, and better estimators require additional computation such as semantic entropy or consistency across samples. This matters for the interface

proposals in Section 6: a visual uncertainty signal is only as trustworthy as the estimator behind it.

### 2.3. Text Simplification: Pedagogical Benefits and Technical Risks

Automatic text simplification has a long history in natural language processing, evolving from rule-based systems applying syntactic transformations and lexical substitution to neural sequence-to-sequence approaches and, more recently, LLMs [5]. Evaluation has traditionally focused on three dimensions: readability (measured by standard formulas such as Flesch Reading Ease [40] and Flesch-Kincaid grade level), semantic preservation (measured by entailment or meaning similarity), and grammaticality or fluency.

Simplification is not information-preserving by default. Work on factuality in simplification shows that models frequently insert errors or drop information, and that the errors are of kinds standard metrics do not catch [31]; Trienes et al. [32] characterise information loss in simplification explicitly and show that it can be surfaced through question-answering. These studies concern general-purpose text (news, encyclopedic and medical content). Our contribution is to move to technical CS concepts, where the information that gets lost tends to be exactly the boundary condition that defines the concept.

Conventional simplification evaluation assumes a monotonic relationship between simplification and information retention: simpler texts should preserve the truth conditions of the original while reducing linguistic complexity. For many domains, including news articles, consumer health information, and government documents, this assumption holds reasonably well. For computer science education, it breaks down. CS concepts are not facts about the world but abstract, relational structures with precise definitions and boundary conditions. A linked list is not merely a chain of connected items, as that description also fits a rope, a queue, or a biological pathway. The pedagogical challenge is to convey both the intuitive metaphor and the precise conditions under which the metaphor fails. Good CS instruction does not replace technical precision with analogy; it layers analogy on top of precision, using metaphor as a scaffold, not a substitute.

Even in the human-authored text literature, easier does not equal better for learning. Coherence-enhancing text revisions helped low-knowledge readers on shallow tests but limited deeper learning for readers with more knowledge [41], and the expertise reversal effect shows that support that helps novices can become a hindrance as knowledge grows [42]. Some difficulty is desirable, because effortful processing supports durable understanding [43]. The implication is not that learners with reading or attention difficulties should face inaccessible text, but that accessibility should reduce extraneous load (decoding, layout, jargon) while leaving intact the germane content (conditions, invariants, exceptions) [44].

LLMs asked to simplify lack this pedagogical awareness. Under pressure to produce simpler output, they can omit boundary conditions, elide exceptions, and promote analogies from the intuitive level to the definitional level. The result is not a simplified version of the same concept but a different, incorrect concept that happens to share surface features with the target. A student who learns that a deadlock is two people stuck in a doorway has not learned a simplified but essentially correct version of deadlock; they have learned a misconception that will later need to be unlearned.

### 2.4. Why Fluent Explanations Mislead: Cognitive Foundations of Vulnerability

The claim that simplified explanations are especially hazardous for learners with reading or attention difficulties rests on four well-established cognitive mechanisms. We present them here as a theoretical motivation; our experiment does not measure them.

Working memory is severely limited [45], and cognitive load theory distinguishes intrinsic, extraneous, and germane load [44]. Automaticity theory holds that when word recognition is not automatic, decoding competes with comprehension for the same limited resources [46]. For a reader with dyslexia, decoding remains effortful longer; for a reader with ADHD, sustaining attention consumes regulation resources [21,22]. Both leave less spare capacity for what a critical reader does on top of comprehension: checking claims against prior knowledge, noticing what is missing, and asking whether an analogy really holds.

Statements that are easy to process are judged more likely to be true and elicit higher confidence [17,18]. A response deliberately engineered to be easy to read therefore also carries a built-in truth cue, whether or not the content deserves it. This is a second-order side effect of accessible design: the same property that reduces decoding cost also lowers the reader's guard.

People typically feel they understand causal systems far better than they can actually explain them, and the feeling is punctured only when they attempt to produce the explanation [19]. A short, vivid, self-contained explanation provides the feeling of understanding without the test, which is the condition under which an incorrect mental model can consolidate without challenge.

Dual-process accounts distinguish fast, effortless, intuitive judgement from slow, effortful checking [47]. Accepting a fluent explanation is the intuitive default; verification is the costly step. Human-automation research shows that people over-rely on automated aids, particularly when verification is expensive, and that interface interventions which force engagement can reduce over-reliance, at the price of extra effort [48]. We revisit this trade-off in Section 6.

Finally, interesting but tangential material can impair learning of the core content [49]. Condition C, which explicitly asks for a memorable and engaging analogy, is exactly the situation in which the vividness of the analogy may compete with, rather than support, the technical content.

### *2.5. Analogies and Misconceptions in Computer Science Education*

Learners do not arrive as blank slates: they build intuitive models, and instruction must confront and restructure them [12,14]. Misconceptions that are consistent with everyday experience are resistant to correction, because a wrong model that successfully explains a subset of observations is unlikely to be abandoned without clear counterexamples [13]. This makes the first exposure to a concept critical.

The study of misconceptions in CS education is well established. Qian and Lehman [15] provide a comprehensive review, identifying common incorrect mental models across introductory programming, such as variable assignment as mathematical equality, the countdown misconception in loops, confusion between reference and value semantics, as well as data structures, such as believing that binary search trees are inherently balanced, and concurrency, such as thinking that race conditions are deterministic or that semaphores prevent all interleavings. Foundational work documented that novice bugs are often language-independent conceptual errors [24], that many misconceptions stem from an inaccurate mental model of the machine [16], and that students' understanding of synchronisation evolves through intermediate, faulty models. These resources provide the ground truth logic for our concept selection and error coding, but they describe misconceptions held by human novices, not those induced by generated explanations.

Analogy is a central instrument of explanation. Structure-mapping theory holds that a good analogy transfers relational structure, not surface attributes [50]. A traffic-light analogy for a semaphore conveys an attribute (something signals go/stop) but not

the relational structure (atomic counting operations coordinating concurrent processes). Analogies are widespread in science and computing teaching, and can improve learning of programming when they match the target model. But it has long been recognised that analogies produce analogy-induced misconceptions, because learners import features of the source that the target does not have [51]. Recommended practice requires explicitly mapping the shared features and, crucially, stating where the analogy breaks down [52]; multiple analogies with complementary strengths are proposed as an antidote to single-analogy misconceptions [51].

Condition C of our study requests one single, vivid analogy, which is exactly the situation the analogy literature warns against: a single analogy, with no stated limits, presented as the core of the explanation. Stone et al. [23] recently extended the analogy analysis to LLM-generated educational content, documenting that common analogies produced by GPT-4 for CS concepts frequently misrepresent technical properties. However, their study used standard prompting without explicit accessibility constraints and did not compare simplification conditions. Our work builds directly on it by systematically varying simplification pressure and measuring the resulting amplification of strategic misconceptions.

## 2.6. LLM-as-Judge Evaluation

Scaling up expert judgement of open-ended text has led to the widespread use of strong LLMs as evaluators. Strong judges can reach agreement with human raters at a level comparable to human-human agreement on chat tasks [33], and rubric-based prompting improves alignment with humans on generation quality [53]. The approach has known biases that shape our design: sensitivity to the position of candidate answers [54], verbosity preferences, and a preference for the judge's own outputs [55]. Section 3 describes how we address self-preference and how we plan to check the judge against human experts; Section 7 discusses the remaining risks.

## 2.7. The Accessibility Paradox: A Formal Hypothesis

Synthesising these streams, we formalise the accessibility paradox as a testable hypothesis. For a given concept  $c$  in computer science, let  $E_s(c)$  be an LLM-generated explanation under a standard prompt (baseline) and  $E_a(c)$  be an explanation under an accessibility prompt requesting simplification for a student with reading or attention difficulties. The strategic misconception rate  $\text{SMR}(E)$  is the probability that  $E$  contains a conceptually coherent but fundamentally incorrect understanding of  $c$ . The oversimplification error rate  $\text{OER}(E)$  is the probability that  $E$  contains an error present in  $E_a$  but absent from  $E_s$  for the same concept.

The accessibility paradox hypothesis states:

$$\text{SMR}(E_a) > \text{SMR}(E_s) \quad \text{and} \quad \text{OER}(E_a) > 0, \quad (1)$$

with the difference increasing as a function of simplification intensity. That is, making explanations more accessible for students with learning difficulties increases, rather than decreases, the rate of strategic misconceptions, and introduces a new class of errors that would not occur in standard explanations.

This hypothesis does not imply that simplification is inherently harmful or that LLMs should not be used as assistive technology. Rather, it predicts a previously undocumented failure mode that current evaluation practices do not detect. The AccessTrap-CS benchmark described in Section 3 was designed to test this hypothesis across models, subfields, and simplification intensities.

### 3. Materials and Methods

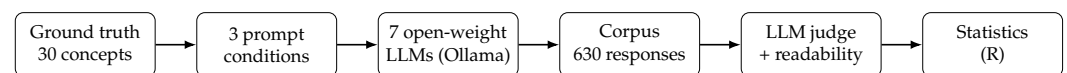
This section describes the benchmark design, concept selection, model selection, prompt engineering, data generation procedure, evaluation framework, and statistical analysis plan. For each design decision we state the rationale, because in a benchmark study the choices are part of the result.

#### 3.1. Benchmark Design Overview

The AccessTrap-CS benchmark tests the accessibility paradox hypothesis across three dimensions: model (7 LLMs), CS subfield (5 subfields), and simplification intensity (3 conditions). The design is fully crossed: each of the 30 concepts is posed to each of the 7 models under each of the 3 conditions, and each cell produces a single response. The 30 concepts and 7 models yield 210 concept-model pairs, each observed under three conditions, so the total corpus size is 630 responses.

$$N_{\text{total}} = C \times M \times K = 30 \times 7 \times 3 = 630 \quad (2)$$

where  $C = 30$  concepts,  $M = 7$  models, and  $K = 3$  conditions. Because the same concept-model pair appears in all three conditions, condition is a within-pair (repeated) factor. This matters for analysis and is what allows the oversimplification error to be defined as a contrast within a pair. Figure 1 shows the overall pipeline.



**Figure 1.** AccessTrap-CS pipeline: from concept ground truth to statistical analysis.

#### 3.2. Concept Selection

A total of 30 core computer science concepts were selected across five subfields: Algorithms and Data Structures, Programming Languages and Paradigms, Database Systems, Computer Networks, and Operating Systems (6 concepts per subfield). The balanced design ensures that subfield comparisons are not driven by unequal sample sizes. Concepts were drawn from standard core undergraduate curricula following the ACM/IEEE-CS/AAAI Computer Science Curricula [56]. Each concept was paired with a ground-truth definition and a reference source from standard textbooks. The ground truth serves as the judge's anchor: without it, the judge would have to rely on its own parametric knowledge, which is exactly the resource whose reliability is under test. The complete list appears in Appendix A.

#### 3.3. Model Selection

Seven open-weight LLMs were selected to ensure reproducibility and to cover a range of model families, parameter scales, and architectural designs. Open weights allow anyone to rerun the whole pipeline locally, without dependence on a vendor's changing API. All models were accessed locally via Ollama running on the Supek HPC cluster.

**Table 1.** Selected open-weight LLMs. Ollama tags and parameter counts were verified with `ollama show`.

| Model (Ollama tag) | Family               | Parameters                               |
|--------------------|----------------------|------------------------------------------|
| mistral:7b         | Mistral AI           | 7B (dense)                               |
| ministral-3:14b    | Mistral AI           | 14B (dense)                              |
| gemma4:31b         | Google               | 31B (dense)                              |
| llama3.3:70b       | Meta                 | 70B (dense)                              |
| deepseek-r1:70b    | DeepSeek (distilled) | 70B (dense, reasoning)                   |
| llama4:16x17b      | Meta                 | 109B total (17B active, 16 experts, MoE) |
| qwen3:235b         | Alibaba              | 235B total (22B active, MoE)             |

Models span dense architectures (7B, 14B, 31B, 70B) and mixture-of-experts (MoE) architectures (llama4:16x17b, qwen3:235b), and include one model with an explicit reasoning-trace output (deepseek-r1:70b). Ollama serves quantised weights by default; we therefore treat model behaviour as that of the quantised builds actually used, and report the exact Ollama tags, quantisation levels, and model digests in the repository. All inference was performed with temperature  $T = 0.7$  and `num_predict = 512` tokens.

The temperature  $T = 0.7$  was chosen to approximate typical chat deployment, where users interact with sampled rather than greedy outputs; the price is run-to-run variability, which we discuss as a limitation in Section 7.

### 3.4. Prompt Engineering and Condition Design

Three prompt templates were designed corresponding to three simplification intensities. All three share the same structure (an instruction followed by `Concept: [CONCEPT]`), so that the only variable is the instruction. The prompts operationalise two recurring accessibility recommendations: reducing decoding load (short sentences, plain vocabulary, no formal notation) and reducing sustained-attention demand (brevity, a single memorable hook).

#### 3.4.1. Condition A: Standard Explanation (Baseline)

The baseline prompt requests an accurate and clear explanation suitable for a first-year computer science student. No simplification constraints are applied. It defines the reference against which errors introduced by simplification are measured.

Explain the following computer science concept accurately and clearly, suitable for a first-year computer science student.

Concept: [CONCEPT]

#### 3.4.2. Condition B: Simplified Explanation (Reading Difficulties)

The simplified prompt instructs the model to address a student with reading difficulties. Constraints include simplified language, short sentences, concrete analogies, avoidance of code and formal definitions, and minimal technical jargon. This condition targets the decoding-load mechanism.

Explain the following computer science concept to a student who has reading difficulties. Use simplified language, short sentences, and concrete analogies instead of code or formal definitions. Avoid technical jargon where possible.

Concept: [CONCEPT]

### 3.4.3. Condition C: Highly Simplified Explanation (Attention Difficulties)

The highly simplified prompt instructs the model to address a student with significant attention difficulties. Constraints include a maximum of five sentences, a single vivid analogy that captures the core idea, and emphasis on memorability and engagement. This condition targets the sustained-attention mechanism and is intended as the higher-intensity condition, because it removes even the space in which qualifications could be placed.

Explain the following computer science concept to a student with significant attention difficulties. Keep the explanation within five sentences. Use one single, vivid analogy that captures the core idea. Make it memorable and engaging.

Concept: [CONCEPT]

### 3.4.4. Construct Validity of the Conditions

Three properties of the prompt design need to be stated explicitly. First, the conditions are bundles of constraints: B and C differ from A, and from each other, in several respects at once (audience label, register, length, number of analogies, instruction to avoid formal definitions). The study therefore tests whether the bundle typical of accessibility-oriented simplification is risky, not the causal contribution of any single element. Second, only condition A contains the word accurately; an accuracy cue is absent in B and C, mirroring how real users write requests but confounding simplification with the absence of that cue. Third, the instruction in B to avoid formal definitions directly discourages the very precision whose absence we count as error. We treat this as the realistic operational form of the paradox (users and teachers do ask for these things) but acknowledge it in Section 7. Finally, the labels dyslexia condition and ADHD condition denote the prompt persona, not the diagnostic status of any real person.

## 3.5. Data Generation Procedure

Data generation was automated using a Python script that iterates over all combinations of concepts, conditions, and models.

### 3.5.1. Input Format

The script reads `ground_truth.csv` (semicolon-delimited) containing the following columns: ID, Domain, Concept, Ground\_Truth, Reference.

### 3.5.2. Generation Loop

For each concept, the script formats the three prompt templates by substituting the concept name. For each of the 7 models, it sends a POST request to the Ollama API at `http://localhost:11434/api/generate`, collects the response text, and appends the response to `corpus.csv` with metadata. A 1-second pause between requests prevents API overload. Failed requests are retried with a 5-second delay. No system prompt, conversation history, or few-shot examples were used, so each response reflects a single-turn interaction.

### 3.5.3. Output Format

The complete corpus `corpus.csv` contains 630 rows with the following schema: `concept_id`, `domain`, `concept`, `condition`, `model`, `prompt`, `response`, `ground_truth`, `reference`, `timestamp`.

### 3.6. Evaluation Framework

Responses were evaluated using an LLM-as-judge approach [33] with Qwen3:235b serving as the expert evaluator. The evaluation script groups responses by concept and model, then presents all three conditions (A, B, C) to the judge simultaneously together with the ground truth. Joint presentation is required for the OER judgement, which is by definition a comparison between a simplified response and the baseline for the same pair; the cost is that the judge's decisions on one condition may be influenced by the others (Section 7).

#### 3.6.1. Judge Model Selection

Qwen3:235b, a 235-billion parameter mixture-of-experts model, was chosen as the judge for three reasons: (a) it is the largest model in the pool and its scale approximates frontier capability, (b) its open-weight license ensures reproducibility, and (c) it is architecturally and organisationally distinct from six of the seven generators. Because Qwen3:235b is also one of the generators, LLM judges are known to favour their own outputs [55]; the evaluation script therefore skips every group whose generating model is Qwen3:235b. As a consequence, 540 of the 630 responses (six generators  $\times$  30 concepts  $\times$  3 conditions) are judged; the 90 Qwen3:235b responses are retained in the corpus and used for the readability analysis only. The judge is configured with temperature  $T = 0.0$  for deterministic outputs and `num_predict = 1024` tokens.

#### 3.6.2. Error Taxonomy

Four evaluation dimensions are defined.

**Strategic Misconception (SM).** A Boolean indicator of whether the explanation contains a subtle but conceptually significant error that sounds plausible but is fundamentally wrong. The defining features are plausibility and consequence: a misstated date or a typographical slip is an error but not a strategic misconception, while claiming that a deadlock is merely processes waiting a long time is.

**Dangerous Analogy (DA).** A Boolean indicator of whether the explanation uses an analogy that is intuitive but misrepresents a key technical property of the concept. An analogy is not dangerous merely for being incomplete; it is dangerous if it invites an inference that is false for the target (e.g., a semaphore as a traffic light invites the inference that a semaphore is a simple binary flag with no atomicity).

**Oversimplification Error (OER).** A Boolean indicator of whether the explanation contains an error that would not appear in a standard explanation, i.e., an error that exists only because of the simplification process. By construction, OER is defined only for conditions B and C, since the reference is condition A.

**Accuracy Score.** A Likert scale from 1 to 5, where 1 = completely wrong and 5 = perfectly accurate. It complements the Boolean indicators by capturing graded degradation that does not cross the threshold of a strategic misconception.

The three Boolean indicators overlap by design (a dangerous analogy is a common route to a strategic misconception, and an oversimplification error is typically also a misconception) but are not identical, and they are analysed both separately and jointly.

#### 3.6.3. Formal Definitions of the Outcome Measures

Let  $r_{cmk}$  denote the response for concept  $c \in \{1, \dots, 30\}$ , model  $m \in \{1, \dots, 7\}$ , and condition  $k \in \{A, B, C\}$ , and let  $SM_{cmk}, DA_{cmk}, OER_{cmk} \in \{0, 1\}$  be the judge's indicators. For a set  $S$  of judged responses,

$$SMR(S) = \frac{1}{|S|} \sum_{(c,m,k) \in S} SM_{cmk}, \quad DAR(S) = \frac{1}{|S|} \sum_{(c,m,k) \in S} DA_{cmk}, \quad (3)$$

and  $OER(S)$  is defined analogously over responses with  $k \in \{B, C\}$ . Condition-level rates are obtained by taking  $S$  to be all judged responses in condition  $k$ ; model- and subfield-level rates by conditioning on  $m$  or on the subfield of  $c$ .

#### 3.6.4. Evaluation Procedure

The evaluation script:

1. Loads `corpus.csv` and groups rows by (`concept_id`, `model`)
2. Skips any group where the generating model equals Qwen3:235b (preventing self-evaluation)
3. For each group, assembles the judge prompt with ground truth and three responses
4. Sends a POST request to the Ollama API
5. Parses the JSON response and flattens it into three rows (one per condition)
6. Appends results to `evaluations.csv`

A 2-second pause between requests prevents API overload. Failed JSON parses are logged for manual inspection. Unrecoverable outputs are excluded from the affected analyses and reported by model and condition, so that missing data are not silently absorbed.

#### 3.6.5. Validation of the Judge

Because all substantive conclusions rest on the judge's classifications, we plan a validation against human experts. A stratified random subsample of 30 concept-model groups (stratified by subfield and model) will be independently annotated by two CS educators blinded to condition and to the judge's output, using the codebook in Appendix C. Agreement between the two humans, and between each human and the judge, will be reported as Cohen's  $\kappa$  [57] interpreted following Landis and Koch [58]. Disagreements will be adjudicated to a consensus label, from which we will estimate the judge's precision and recall for each Boolean indicator.

#### 3.7. Readability Metrics

As a manipulation check, all responses are analysed for quantitative readability using three metrics computed with the `textstat` Python library: Flesch Reading Ease (higher = easier) [40], Flesch-Kincaid Grade Level, and average sentence length in words:

$$FRE = 206.835 - 1.015 \frac{\text{words}}{\text{sentences}} - 84.6 \frac{\text{syllables}}{\text{words}}, \quad (4)$$

$$FKGL = 0.39 \frac{\text{words}}{\text{sentences}} + 11.8 \frac{\text{syllables}}{\text{words}} - 15.59. \quad (5)$$

These formulas depend only on sentence and syllable length, i.e., surface form. They are appropriate for verifying that the prompts changed the text as intended, but they do not measure comprehensibility, and technical terms and code-like tokens can distort syllable counts. Readability is therefore used as a manipulation check and not as an outcome of interest.

#### 3.8. Statistical Analysis Plan

All statistical tests will be performed in R (version 4.3.2) [59]. The significance threshold is set at  $\alpha = 0.05$  with adjustments for multiple comparisons where indicated.

##### 3.8.1. Primary Hypothesis Testing

The primary hypothesis  $SMR(E_a) > SMR(E_s)$  is tested using a one-way analysis of variance (ANOVA) with condition (A\_standard, B\_simplified, C\_highly\_simplified) as the independent variable and strategic misconception rate as the dependent variable. For

significant ANOVA results ( $p < 0.05$ ), post-hoc pairwise comparisons are conducted using Tukey's Honestly Significant Difference (HSD) test [60].

Normality of residuals is assessed using Shapiro-Wilk tests [61]; homogeneity of variances is assessed using Levene's test [62]. Given the large sample size ( $n = 630$ ), ANOVA is robust to modest violations of normality.

### 3.8.2. Robustness Analyses for Binary, Paired Outcomes

SM, DA, and OER are binary indicators, and the three conditions for the same concept-model pair are not independent observations. ANOVA on binary outcomes is a linear probability model, and treating repeated observations as independent tends to overstate precision. We therefore complement the primary analysis with methods that respect these features:

- Cochran's Q test [63] across the three conditions with concept-model pair as the blocking unit, followed by pairwise McNemar tests [64] with Holm correction;
- a mixed-effects logistic regression [65],

$$\text{logit Pr}(\text{SM}_{cmk} = 1) = \beta_0 + \beta_B \mathbb{I}[k=B] + \beta_C \mathbb{I}[k=C] + u_c + v_m, \quad (6)$$

with random intercepts  $u_c$  for concept and  $v_m$  for model (and, in extended models, a condition  $\times$  model interaction), reporting odds ratios with 95% confidence intervals;

- for the ordinal accuracy score, a non-parametric alternative (Friedman or cumulative-link mixed model) alongside ANOVA.

Where the robustness analyses and the ANOVA disagree, we report both and interpret the more conservative.

### 3.8.3. Secondary Analyses

Chi-square tests of independence examine associations between condition and (a) dangerous analogy presence, (b) analogy type, and (c) specific error subtypes. Two-way ANOVA with model and condition as factors compares model families. One-way ANOVA within each condition examines subfield differences. Across the family of secondary tests, the false discovery rate is controlled using the Benjamini-Hochberg procedure [66].

### 3.8.4. Effect Sizes

Effect sizes are reported as Cohen's  $d$  for pairwise comparisons,  $\eta^2$  for ANOVA, and Cramér's  $V$  for chi-square tests [67,68], with

$$V = \sqrt{\frac{\chi^2}{N(\min(r, c) - 1)}}. \quad (7)$$

Effect sizes are interpreted alongside confidence intervals, since with a corpus of this size small differences may be statistically significant without being educationally meaningful.

## 3.9. Reproducibility

The complete AccessTrap-CS corpus, all prompt templates, evaluation codebook, generation and evaluation scripts, and analysis code are publicly available at [repository URL]. Model inference was performed with Ollama on the Supek HPC cluster; the Ollama version, model digests, and sampling parameters are logged with the corpus.

4. Results

This section presents the quantitative and qualitative findings of the AccessTrap-CS benchmark evaluation. The corpus contains 630 responses generated across 30 computer science concepts, three simplification conditions, and seven open-weight LLMs. Readability analyses use all 630 responses; analyses of SM, DA, OER, and accuracy use the 540 responses judged by Qwen3:235b (six generating models). Sample sizes for every table will be stated explicitly.

- 4.1. Readability Metrics
- 4.2. Strategic Misconception Rate (SMR)
- 4.3. Dangerous Analogy Rate (DAR)
- 4.4. Oversimplification Error Rate (OER)
- 4.5. Accuracy Scores
- 4.6. Subfield Analysis
- 4.7. Model Comparison
- 4.8. Judge Validation
- 4.9. Summary of Findings

5. Discussion

This section interprets the findings of the AccessTrap-CS benchmark in the context of the accessibility paradox hypothesis, existing literature on misconceptions in computer science education, and the design of AI tutoring systems.

- 5.1. Testing the Accessibility Paradox
- 5.2. Why Vulnerable Learners Are Disproportionately at Risk

The interpretation below is theoretical and is offered as a framework for reading the results; our experiment does not measure learner cognition (see Section 7). The first reason is asymmetry of verification. A simplified explanation is not merely less detailed; it removes the cues a reader would use to notice that something is missing. Errors of omission are invisible to the person who does not already possess the omitted content, and novices lack the schemas that would flag the gap [42]. The second is load. If decoding or attention regulation already absorbs a large share of working-memory resources [44–46], then the additional operation of checking a claim against prior knowledge competes with comprehension itself, and the reader is likely to fall back on the cheap intuitive judgement [47]. The third is that simplification makes the wrong explanation more persuasive, not less: fluent statements gain a truth premium [17,18], and a short self-contained account produces a feeling of understanding that is not tested [19]. The fourth is the role of analogy. A vivid analogy is memorable by design, so a flawed one is also memorably flawed; it can also function as a seductive detail that captures attention at the expense of the technical core [49], and a single analogy without a stated limit is the textbook recipe for analogy-induced misconception [51,52]. Finally, the deployment context removes external checks: the tutor is private, and there is often no teacher, peer, or textbook running in parallel.

It is important to state what this argument does not claim. It does not claim that learners with dyslexia or ADHD are less capable of critical thought. The vulnerability is situational and produced jointly by the learner’s load and the properties of the material; the same reasoning predicts risk for any novice reading under time pressure or fatigue. It also does not claim that simplification should be abandoned. The literature on dual-process reasoning is a framework for organising these considerations rather than a validated model of the specific mechanism, and only a study with learners can establish how strongly each factor contributes.

### 5.3. *The Systematic Nature of Oversimplification Errors*

The theoretical argument for why oversimplification errors should be systematic rather than random runs as follows. A CS concept is defined by a small set of properties, several of which are relational or conditional (invariants, necessary conditions, boundary cases). An analogy typically maps some attributes of a familiar situation onto the target but not its relational structure [50]. When a prompt fixes a strong compression budget (short sentences, five sentences in total, a single analogy), the model has to choose which properties to retain, and the ones that are easiest to render in everyday language are the ones that survive. Necessary conditions (circular wait, atomic) and quantifiers (only if, at least) are the first to be dropped, because they are the least concrete. The resulting error is thus predictable from the structure of the concept and the size of the compression budget, which is why we regard it as systematic.

We anticipate several recurring subtypes, which will be derived inductively from the data rather than imposed in advance: (i) omission of a necessary condition; (ii) promotion of an analogy from illustration to definition; (iii) over-generalisation of a property that holds only in a special case; (iv) conflation with a neighbouring concept; (v) replacement of a mechanism by an outcome. Concept-specific results, including which concepts and subfields prove most susceptible, will be added once the analysis is complete.

The long-term consequences are amplified by the hierarchical nature of CS knowledge: later topics (scheduling, synchronisation, transactions, congestion control) build on earlier ones [24], and, once formed, an intuitively appealing wrong model resists later instruction [12,13]. An oversimplification error at the first exposure may therefore become a persistent obstacle that must later be unlearned.

### 5.4. *Implications for AI Safety in Education*

Three implications follow from the design of the study, independently of the specific numerical results. First, accuracy on a neutral prompt is an insufficient safety criterion for an educational assistant. Models must be evaluated conditioned on the audience and instruction they will actually receive, and benchmark suites should include persona- and format-conditioned variants [29]. Second, standard quality metrics (readability, fluency, coherence) can be optimised in ways that mask semantic degradation: a response can improve on every conventional measure of accessibility while getting worse on conceptual fidelity [31,32]. This suggests that simplification evaluation should include a fidelity check against a list of necessary properties of the concept. Third, as with warmth [25], the pressure to please and the pressure to simplify are pressures on the optimisation target, not defects of any particular model, so they are unlikely to be resolved by scale alone [29].

Mitigation can be pursued at two levels. At the model or prompt level, simplification instructions can be rewritten so that boundary conditions are preserved (e.g., requiring an explicit statement of where each analogy fails, in line with the teaching-with-analogies model [52]), and generate-then-verify pipelines can compare a simplified explanation with a ground-truth checklist. At the interface level, uncertainty and analogy status can be surfaced to the learner, which is the subject of Section 6. Neither level is evaluated in this paper; we present them as hypotheses that follow from the mechanism above.

## 6. **Design Principles for Accessible AI Tutoring Interfaces**

The findings of this study have direct implications for the design of AI tutoring interfaces, particularly for students with learning difficulties. Rather than relying exclusively on text-based disclaimers (e.g., AI can make mistakes), which vulnerable learners may overlook or ignore, we propose three accessibility-first design principles that leverage visual signalling to communicate model uncertainty without increasing cognitive load. The

principles are derived from the mechanism analysis in Section 5 and from established work on accessible design, and they are proposals: their effectiveness for the target population has not been tested here.

The principles draw on three bodies of work. Universal Design for Learning and ability-based design argue that systems should adapt to the learner's abilities and provide multiple means of representation instead of adding remedial layers [69,70]. Research on visual perception shows that certain features (colour, shape, orientation, motion) are detected pre-attentively, before effortful reading begins [71]. Research on uncertainty communication shows that visualisation of uncertainty can be effective but requires careful evaluation, and that how a model expresses uncertainty changes users' reliance and trust [72].

### 6.1. Principle 1: Visual Uncertainty Signalling

Standard text-based disclaimers are cognitively opaque: they occupy prominent screen real estate but are easily dismissed as generic boilerplate. For students with reading difficulties or attention challenges, such disclaimers represent additional text to decode rather than actionable information. We propose replacing or augmenting text disclaimers with visual uncertainty indicators that can be processed pre-attentively [71].

- Confidence colour coding. Responses generated with low model confidence are displayed with a subtle amber background. High-confidence responses use a neutral or green background. Colour must not be the only channel (see Principle 3); it is paired with the icon below.
- Uncertainty icons. A small, standardised icon (e.g., a dotted outline or question mark) appears adjacent to responses containing high uncertainty. The icon persists even when the user scrolls, providing continuous awareness of reliability.
- Progressive disclosure of uncertainty details. Tapping or hovering over the uncertainty indicator reveals a plain-language explanation. This design places uncertainty information at the point of need without overwhelming the primary interface.

Caveat on the signal source. The interface can only convey what the system can estimate. Verbalised model confidence is often poorly calibrated, and better indicators require additional computation such as semantic entropy or consistency across samples. A miscalibrated indicator is arguably worse than none, because it produces unwarranted trust. An implementation should therefore treat the estimator's validation as part of the design.

### 6.2. Principle 2: Analogy Status Tagging

Dangerous analogies are a plausible contributor to strategic misconceptions, yet analogies remain a critical accessibility tool for making abstract concepts concrete. Rather than eliminating analogies, interfaces should tag them explicitly, distinguishing between pedagogical analogies (acknowledged as simplifications) and false equivalences (presented as literal descriptions). This operationalises the central recommendation of the teaching-with-analogies model, that the learner be told where the analogy holds and where it breaks [52].

- Analogy badges. Each analogy in the generated response is identified and tagged with a small badge reading Analogy or Simplification. The badge serves as a metacognitive prompt, reminding the learner that the statement is a comparison, not a definition.
- Toggle between analogy and technical description. For responses containing analogies, users can toggle to a technical version that replaces the analogy with precise terminology.

- Analogy boundary highlighting. The specific sentence or clause containing the analogy is highlighted with a dashed border, making visually explicit where the model has shifted from technical description to figurative comparison.
- Limits-of-analogy statement. Each analogy is accompanied, on demand, by a one-line statement of the main property it does not share with the target (e.g., a semaphore also has a counter and cannot be interrupted midway), and, where useful, a second analogy that highlights a complementary aspect [51].

A practical route to badges and boundaries is to require the model to return structured output in which analogy spans are marked; the reliability of such self-tagging is an empirical question.

### 6.3. Principle 3: Accessibility-First Without Text-Based Disclaimers

Current AI interfaces often default to text-based safety disclaimers that are inaccessible to the very populations they aim to protect. Students with dyslexia may skip long disclaimers entirely; students with ADHD may lose focus before reaching the disclaimer. The proposed principles invert this logic: safety information should be integrated into the primary visual channel rather than relegated to peripheral text.

- No mandatory text disclaimers. Text-only disclaimers are neither necessary nor sufficient for accessibility. When they must be included (e.g., for regulatory compliance), they should be collapsible with a visual indicator of their collapsed state.
- Visual-first, not visual-only, communication. Critical safety information should be communicable through colour, shape, and iconography, with text as a secondary channel for users who opt to explore details. Every visual cue must also have a programmatic text alternative, because many learners with dyslexia use text-to-speech or screen readers, and because colour must not be the sole means of conveying information [73].
- Restrained motion. Animation may be offered to attract attention to a signal, but it is optional and off by default under a reduced-motion preference, since moving elements can themselves be distracting for learners with attention difficulties.
- Customisable visual themes. Users should be able to select visual themes optimised for their specific needs (e.g., high-contrast mode for dyslexia, reduced motion for attention difficulties) without losing uncertainty signalling [69,70].

These principles shift responsibility from the learner (who must remember to be sceptical) to the interface (which designs scepticism into the visual language). For students with reduced spare capacity for parallel fact-verification, this shift is plausibly not merely a convenience but a prerequisite for safe AI-assisted learning.

### 6.4. Tensions and Evaluation Roadmap

Two tensions should be acknowledged. Interventions that force the learner to engage (cognitive forcing) reduce over-reliance but increase effort, and are less liked [48]; for the target population that additional effort has a cost, so the principles favour passive, glanceable signals over forcing functions. In addition, adding indicators adds interface elements and therefore load, so the signals must be few, consistent, and learnable [44]. Evaluating the principles requires a study with learners with and without dyslexia and ADHD, comparing interfaces with and without the signals on measurable outcomes: acceptance of a deliberately flawed explanation, later unassisted performance (following the logic of [11]), perceived workload, and usability. We propose this as future work.

7. Limitations

This study has several limitations that should be considered when interpreting the findings.

7.1. Generalisation to Other Populations

The evaluation in this study was performed by an LLM judge, not by students with learning difficulties. While this design provides scalable and reproducible evaluation, it does not measure actual student outcomes. Whether strategic misconceptions produce measurable learning deficits in dyslexic or ADHD students, and whether the plausibility of a flawed explanation predicts actual student belief change, remain open empirical questions. The mechanisms in Section 5 are theoretical. Moreover, the dyslexia and ADHD conditions are prompt personas: they simulate an interaction style and cannot represent the heterogeneity of real learners, many of whom will ask for different kinds of support. Future work should replicate this study with target population samples using learning outcome measures.

7.2. Generalisation to Other Domains

The benchmark is restricted to computer science education, and to English-language prompts. Other STEM domains (e.g., physics, mathematics) and non-STEM fields (e.g., history, law) may exhibit different simplification dynamics. For example, fields with higher tolerance for interpretive variation may show lower rates of oversimplification errors, while fields with axiomatic knowledge structures may show higher rates. Extending AccessTrap-CS to additional disciplines and languages is a priority for future work.

7.3. Generalisation to Other Models

The seven evaluated models, while diverse in scale and architecture, represent only a fraction of available LLMs. Proprietary models such as GPT-4o, Claude 3.5, and Gemini 1.5 were not included due to their closed-weight nature and API costs. It remains unknown whether the pattern observed here generalises to these systems, which may incorporate additional safety fine-tuning. Ollama serves quantised weights by default, so behaviour may differ from full-precision deployments.

7.4. Prompt Design and Construct Validity

As explained in Section 3, the conditions are bundles of constraints; the accuracy cue appears only in condition A; and condition B explicitly discourages formal definitions. The results therefore describe the behaviour of models under realistic accessibility-style requests, not the causal effect of any single element such as analogy use or length limit. Ablation prompts (e.g., simplification with an explicit accuracy instruction, or a single-analogy requirement without an audience label) would separate these factors. Each cell also contains a single generation per concept-model-condition, so the estimates reflect between-concept and between-model variability but not within-cell sampling variability.

7.5. Causal Interpretation of Oversimplification Errors

The classification of oversimplification errors relies on the assumption that an error present in a simplified response but absent from the baseline response for the same concept and model was caused by the simplification pressure. While this design controls for concept and model variation, it does not control for random variation in model outputs. Because generation was performed with temperature  $T = 0.7$ , two generations from the same model under identical conditions could produce different outputs, so an error attributed to simplification might have appeared in a different baseline sample. Replication with

temperature  $T = 0.0$ , or with several samples per cell, would isolate the simplification effect but would also reduce ecological validity.

7.6. *LLM-as-Judge Validation*

Qwen3:235b was used as the primary evaluator. While this model is the largest in our set and its open-weight license ensures reproducibility, the validation of its judgments against human experts is limited. Judges show position, verbosity, and self-preference biases [33,54,55]; we mitigate the last by not judging Qwen3’s own outputs, which however leaves 90 responses unjudged and makes the model comparison unbalanced. Joint presentation of the three conditions to the judge can create contrast effects, in which the judge’s rating of one response depends on the others. The ground truth is a single reference definition per concept, which may not capture legitimate alternative formulations and could lead to false positives. Future work should conduct comprehensive human-LLM agreement analysis across the full distribution of responses.

7.7. *Parsing Failures and Output Length*

DeepSeek-R1 exhibited parsing issues due to its unique output format, which wraps reasoning traces within <think> tags. In some instances, these tags interfered with programmatic segmentation, resulting in unrecoverable text segments. Because reasoning tokens count toward the generation limit, a cap of 512 tokens may also truncate the visible answer of reasoning models. This limitation is specific to reasoning-style output and does not affect the other models, but it highlights a broader challenge: as models evolve, they may adopt output formats that break standard parsing assumptions. Future versions of the benchmark should implement model-specific parsers and length budgets.

8. **Conclusions**

This paper introduced the accessibility paradox: the hypothesis that LLM simplification for students with learning difficulties, while well-intentioned, may systematically increase the rate of strategic misconceptions. We presented AccessTrap-CS, a benchmark corpus of 630 responses generated by seven open-weight LLMs across 30 core computer science concepts under three simplification conditions, together with an error taxonomy comprising strategic misconceptions, dangerous analogies, and oversimplification errors, and an LLM-as-judge evaluation with Qwen3:235b as the evaluator.

[Summary of empirical findings to be inserted after analysis.]

Beyond empirical findings, this paper contributes three accessibility-first design principles for AI tutoring interfaces: visual uncertainty signalling, analogy status tagging, and visual-first communication that does not rely on text-based disclaimers. These principles shift responsibility for scepticism from the learner to the interface, a design move that is plausibly important for students with reduced spare capacity for parallel fact-verification.

Future work should extend AccessTrap-CS to additional domains and languages, add ablation prompts and multiple samples per cell, replicate the study with target population samples using learning outcome measures, and implement and evaluate the proposed design principles in tutoring systems in real-world settings. As LLMs become increasingly integrated into educational practice, the ability to detect, measure, and mitigate strategic misconceptions will be essential for ensuring that accessibility tools do not inadvertently become vectors of conceptual harm.

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**Data Availability Statement:** The AccessTrap-CS corpus and evaluation scripts are available at [repository URL to be added upon acceptance].

**Conflicts of Interest:** The authors declare no conflicts of interest.

**Abbreviations**

The following abbreviations are used in this manuscript:

|       |                                          |
|-------|------------------------------------------|
| LLM   | Large Language Model                     |
| CS    | Computer Science                         |
| SMR   | Strategic Misconception Rate             |
| DAR   | Dangerous Analogy Rate                   |
| OER   | Oversimplification Error Rate            |
| ADHD  | Attention Deficit Hyperactivity Disorder |
| ANOVA | Analysis of Variance                     |
| HSD   | Honestly Significant Difference          |
| HPC   | High Performance Computing               |
| MoE   | Mixture of Experts                       |
| FRE   | Flesch Reading Ease                      |
| FKGL  | Flesch-Kincaid Grade Level               |

**Appendix A CS Concepts and Subfields**

**Appendix B Prompt Templates**

The three templates used in the study are reproduced verbatim below. [CONCEPT] is replaced by the concept name.

**Condition A (standard).**

Explain the following computer science concept accurately and clearly, suitable for a first-year computer science student.

Concept: [CONCEPT]

**Condition B (simplified, reading difficulties).**

Explain the following computer science concept to a student who has reading difficulties. Use simplified language, short sentences, and concrete analogies instead of code or formal definitions. Avoid technical jargon where possible.

Concept: [CONCEPT]

**Condition C (highly simplified, attention difficulties).**

Explain the following computer science concept to a student with significant attention difficulties. Keep the explanation within five sentences. Use one single, vivid analogy that captures the core idea. Make it memorable and engaging.

Concept: [CONCEPT]

## Appendix C Evaluation Codebook

## Appendix D Annotated Examples of Dangerous Analogies

### References

1. Zdravkova, K.; Krasniqi, V.; Dalipi, F.; Ferati, M. Cutting-edge communication and learning assistive technologies for disabled children: An artificial intelligence perspective. *Frontiers in Artificial Intelligence* **2022**, *5*, 970430. <https://doi.org/10.3389/frai.2022.970430>.
2. Kortemeyer, G. Could an artificial-intelligence agent pass an introductory physics course? *Physical Review Physics Education Research* **2023**, *19*, 010132. <https://doi.org/10.1103/PhysRevPhysEducRes.19.010132>.
3. Kasneci, E.; Seßler, K.; Küchemann, S.; Bannert, M.; Dementieva, D.; et al. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences* **2023**, *103*, 102274. <https://doi.org/10.1016/j.lindif.2023.102274>.
4. Rello, L.; Baeza-Yates, R. Good fonts for dyslexia. In Proceedings of the Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility, 2013, pp. 1–8. <https://doi.org/10.1145/2513383.2513447>.
5. Štajner, S.; Saggion, H. Automatic text simplification for reading assistance: A review of methods, applications, and future directions. *Natural Language Engineering* **2021**, *27*, 507–546. <https://doi.org/10.1017/S1351324920000525>.
6. Finnie-Ansley, J.; Denny, P.; Becker, B.A.; Luxton-Reilly, A.; Prather, J. The robots are coming: Exploring the implications of OpenAI Codex on introductory programming. In Proceedings of the Proceedings of the 24th Australasian Computing Education Conference, 2022, pp. 10–19. <https://doi.org/10.1145/3511861.3511863>.
7. Prather, J.; Denny, P.; Leinonen, J.; Becker, B.A.; Albluwi, I.; et al. The robots are here: Navigating the generative AI revolution in computing education. In Proceedings of the Proceedings of the Working Group Reports on Innovation and Technology in Computer Science Education, 2023, pp. 97–149. <https://doi.org/10.1145/3623762.3633499>.
8. Denny, P.; Prather, J.; Becker, B.A.; Finnie-Ansley, J.; Hellas, A.; et al. Computing education in the era of generative AI. In Proceedings of the Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 2024. <https://doi.org/10.1145/3626252>.
9. Pardos, Z.A.; Bhandari, S. ChatGPT-generated help produces learning gains equivalent to human tutor-authored help on mathematics skills. *PLOS ONE* **2024**, *19*, e0304013. <https://doi.org/10.1371/journal.pone.0304013>.
10. Kestin, G.; Miller, K.; Klaes, A.; Milbourne, T.; Ponti, G. AI tutoring outperforms in-class active learning: An RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports* **2025**, *15*, 17458. <https://doi.org/10.1038/s41598-025-97652-6>.
11. Bastani, H.; Bastani, O.; Sungu, A.; Ge, H.; Kabakçı, Ö.; Mariman, R. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences* **2025**, *122*, e2422633122. <https://doi.org/10.1073/pnas.2422633122>.
12. Posner, G.J.; Strike, K.A.; Hewson, P.W.; Gertzog, W.A. Accommodation of a scientific conception: Toward a theory of conceptual change. *Science Education* **1982**, *66*, 211–227. <https://doi.org/10.1002/sce.3730660207>.
13. Chi, M.T.H.; Slotta, J.D.; de Leeuw, N. From things to processes: A theory of conceptual change for learning science concepts. *Learning and Instruction* **1994**, *4*, 27–43. [https://doi.org/10.1016/0959-4752\(94\)90017-5](https://doi.org/10.1016/0959-4752(94)90017-5).
14. Vosniadou, S.; Brewer, W.F. Mental models of the earth: A study of conceptual change in childhood. *Cognitive Psychology* **1992**, *24*, 535–585. [https://doi.org/10.1016/0010-0285\(92\)90018-W](https://doi.org/10.1016/0010-0285(92)90018-W).
15. Qian, Y.; Lehman, J.D. Students' misconceptions and other difficulties in introductory programming: A literature review. *ACM Transactions on Computing Education* **2017**, *18*, 1–24. <https://doi.org/10.1145/3077618>.
16. Sorva, J. Visual Program Simulation in Introductory Programming Education. PhD thesis, Aalto University, 2012.
17. Reber, R.; Schwarz, N. Effects of perceptual fluency on judgments of truth. *Consciousness and Cognition* **1999**, *8*, 338–342. <https://doi.org/10.1006/ccog.1999.0392>.
18. Alter, A.L.; Oppenheimer, D.M. Uniting the tribes of fluency to form a metacognitive nation. *Personality and Social Psychology Review* **2009**, *13*, 219–235. <https://doi.org/10.1177/1088868309341564>.
19. Rozenblit, L.; Keil, F. The misunderstood limits of folk science: An illusion of explanatory depth. *Cognitive Science* **2002**, *26*, 521–562. [https://doi.org/10.1207/s15516709cog2605\\_1](https://doi.org/10.1207/s15516709cog2605_1).
20. Snowling, M.J.; Hulme, C.; Nation, K. Defining and understanding dyslexia: past, present and future. *Oxford Review of Education* **2019**, *46*, 501–513. <https://doi.org/10.1080/03054985.2020.1765756>.
21. Barkley, R.A. *Attention-Deficit Hyperactivity Disorder: A Handbook for Diagnosis and Treatment*, 4th ed.; Guilford Publications, 2015.
22. Martinussen, R.; Hayden, J.; Hogg-Johnson, S.; Tannock, R. A meta-analysis of working memory impairments in children with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child & Adolescent Psychiatry* **2005**, *44*, 377–384. <https://doi.org/10.1097/01.chi.0000153228.72591.73>.

23. Stone, M.; Liao, Y.; Szafir, D.; Boyer, K.E. Analogies in the age of LLMs: When explanatory metaphors mislead CS learners. In Proceedings of the Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 2024, pp. 1256–1262. <https://doi.org/10.1145/3626253.3635473>.
24. Ben-Ari, M. Constructivism in computer science education. *Journal of Computers in Mathematics and Science Teaching* **2001**, *20*, 45–73.
25. Ibrahim, H.; Liu, T.; Chen, A.; Wang, D. The warmth-accuracy trade-off: When making LLMs more empathetic increases factual errors. In Proceedings of the Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency, 2026. Forthcoming.
26. Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C.; et al. Training language models to follow instructions with human feedback. In Proceedings of the Advances in Neural Information Processing Systems, 2022, Vol. 35, pp. 27730–27744. <https://doi.org/10.48550/arXiv.2203.02155>.
27. Sharma, M.; Tong, M.; Korbak, T.; Duvenaud, D.; Askell, A.; et al. Towards understanding sycophancy in language models. In Proceedings of the Proceedings of the 12th International Conference on Learning Representations, 2024. <https://doi.org/10.48550/arXiv.2310.13548>.
28. Perez, E.; Ringer, S.; Lukošiušė, K.; Nguyen, K.; Chen, E.; et al. Discovering language model behaviors with model-written evaluations. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2023, 2023, pp. 13387–13434. <https://doi.org/10.18653/v1/2023.findings-acl.847>.
29. Lin, S.; Hilton, J.; Evans, O. TruthfulQA: Measuring how models mimic human falsehoods. In Proceedings of the Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 3214–3252. <https://doi.org/10.18653/v1/2022.acl-long.229>.
30. Gadiraju, V.; Kane, S.; Dev, S.; Taylor, A.; Wang, D.; Denton, E.; Brewer, R. "I wouldn't say offensive but...": Disability-centered perspectives on large language models. In Proceedings of the Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency, 2023, pp. 205–216. <https://doi.org/10.1145/3593013.3593989>.
31. Devaraj, A.; Sheffield, W.; Wallace, B.C.; Li, J.J. Evaluating factuality in text simplification. In Proceedings of the Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 7331–7345. <https://doi.org/10.18653/v1/2022.acl-long.505>.
32. Trienes, J.; Joseph, S.; Schlötterer, J.; Seifert, C.; Lo, K.; Xu, W.; Wallace, B.C.; Li, J.J. InfoLossQA: Characterizing and recovering information loss in text simplification. In Proceedings of the Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024, pp. 4263–4294. <https://doi.org/10.18653/v1/2024.acl-long.234>.
33. Zheng, L.; Chiang, W.L.; Sheng, Y.; Zhuang, S.; Wu, Z.; et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In Proceedings of the Advances in Neural Information Processing Systems, 2023, Vol. 36. <https://doi.org/10.48550/arXiv.2306.05685>.
34. Lyon, G.R.; Shaywitz, S.E.; Shaywitz, B.A. A definition of dyslexia. *Annals of Dyslexia* **2003**, *53*, 1–14. <https://doi.org/10.1007/s11881-003-0001-9>.
35. Faraone, S.V.; Asherson, P.; Banaschewski, T.; Biederman, J.; Buitelaar, J.K.; et al. Attention-deficit/hyperactivity disorder. *Nature Reviews Disease Primers* **2015**, *1*, 15020. <https://doi.org/10.1038/nrdp.2015.20>.
36. Peng, P.; Fuchs, D. A meta-analysis of working memory deficits in children with learning difficulties. *Journal of Learning Disabilities* **2016**, *49*, 3–20. <https://doi.org/10.1177/0022219414528532>.
37. Rello, L.; Baeza-Yates, R.; Bott, S.; Saggion, H. Simplify or help? Text simplification strategies for people with dyslexia. In Proceedings of the Proceedings of the 10th International Cross-Disciplinary Conference on Web Accessibility, 2013, pp. 1–10. <https://doi.org/10.1145/2461121.2461126>.
38. Ji, Z.; Lee, N.; Frieske, R.; Yu, T.; Su, D.; et al. Survey of hallucination in natural language generation. *ACM Computing Surveys* **2023**, *55*, 1–38. <https://doi.org/10.1145/3571730>.
39. Huang, L.; Yu, W.; Ma, W.; Zhong, W.; Feng, Z.; et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems* **2025**, *43*, 1–55. <https://doi.org/10.1145/3703155>.
40. Flesch, R. A new readability yardstick. *Journal of Applied Psychology* **1948**, *32*, 221–233. <https://doi.org/10.1037/h0057532>.
41. McNamara, D.S.; Kintsch, E.; Songer, N.B.; Kintsch, W. Are good texts always better? Interactions of text coherence, background knowledge, and levels of understanding in learning from text. *Cognition and Instruction* **1996**, *14*, 1–43. [https://doi.org/10.1207/s1532690xc1401\\_1](https://doi.org/10.1207/s1532690xc1401_1).
42. Kalyuga, S.; Ayres, P.; Chandler, P.; Sweller, J. The expertise reversal effect. *Educational Psychologist* **2003**, *38*, 23–31. [https://doi.org/10.1207/S15326985EP3801\\_4](https://doi.org/10.1207/S15326985EP3801_4).
43. Bjork, E.L.; Bjork, R.A. Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In *Psychology and the Real World: Essays Illustrating Fundamental Contributions to Society*; Gernsbacher, M.A.; Pew, R.W.; Hough, L.M.; Pomerantz, J.R., Eds.; Worth Publishers, 2011; pp. 56–64.
44. Sweller, J.; van Merriënboer, J.J.G.; Paas, F.G.W.C. Cognitive architecture and instructional design. *Educational Psychology Review* **1998**, *10*, 251–296. <https://doi.org/10.1023/A:1022193728205>.

45. Baddeley, A.D.; Hitch, G. Working memory. *Psychology of Learning and Motivation* **1974**, *8*, 47–89. [https://doi.org/10.1016/S0079-7421\(08\)60452-1](https://doi.org/10.1016/S0079-7421(08)60452-1). 1026
46. LaBerge, D.; Samuels, S.J. Toward a theory of automatic information processing in reading. *Cognitive Psychology* **1974**, *6*, 293–323. [https://doi.org/10.1016/0010-0285\(74\)90015-2](https://doi.org/10.1016/0010-0285(74)90015-2). 1027
47. Kahneman, D. *Thinking, Fast and Slow*; Farrar, Straus and Giroux, 2011. 1028
48. Buçinca, Z.; Malaya, M.B.; Gajos, K.Z. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction* **2021**, *5*, 1–21. <https://doi.org/10.1145/3449282>. 1029
49. Harp, S.F.; Mayer, R.E. How seductive details do their damage: A theory of cognitive interest in science learning. *Journal of Educational Psychology* **1998**, *90*, 414–434. <https://doi.org/10.1037/0022-0663.90.3.414>. 1030
50. Gentner, D. Structure-mapping: A theoretical framework for analogy. *Cognitive Science* **1983**, *7*, 155–170. [https://doi.org/10.1207/s15516709cog0702\\_3](https://doi.org/10.1207/s15516709cog0702_3). 1031
51. Spiro, R.J.; Feltovich, P.J.; Coulson, R.L.; Anderson, D.K. Multiple analogies for complex concepts: Antidotes for analogy-induced misconception in advanced knowledge acquisition. In *Similarity and Analogical Reasoning*; Vosniadou, S.; Ortony, A., Eds.; Cambridge University Press, 1989; pp. 498–531. <https://doi.org/10.1017/CBO9780511529863.023>. 1032
52. Glynn, S.M. Explaining science concepts: A teaching-with-analogies model. In *The Psychology of Learning Science*; Glynn, S.M.; Yeany, R.H.; Britton, B.K., Eds.; Lawrence Erlbaum Associates, 1991; pp. 219–240. 1033
53. Liu, Y.; Iter, D.; Xu, Y.; Wang, S.; Xu, R.; Zhu, C. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 2511–2522. <https://doi.org/10.18653/v1/2023.emnlp-main.153>. 1034
54. Wang, P.; Li, L.; Chen, L.; Cai, Z.; Zhu, D.; et al. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926* **2023**. <https://doi.org/10.48550/arXiv.2305.17926>. 1035
55. Panickssery, A.; Bowman, S.R.; Feng, S. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076* **2024**. <https://doi.org/10.48550/arXiv.2404.13076>. 1036
56. ACM/IEEE-CS/AAAI Joint Task Force on Computing Curricula. *Computer Science Curricula 2023*, 2023. 1037
57. Cohen, J. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **1960**, *20*, 37–46. <https://doi.org/10.1177/001316446002000104>. 1038
58. Landis, J.R.; Koch, G.G. The measurement of observer agreement for categorical data. *Biometrics* **1977**, *33*, 159–174. <https://doi.org/10.2307/2529310>. 1039
59. R Core Team. *R: A Language and Environment for Statistical Computing*, 2023. 1040
60. Tukey, J.W. Comparing individual means in the analysis of variance. *Biometrics* **1949**, *5*, 99–114. <https://doi.org/10.2307/3001913>. 1041
61. Shapiro, S.S.; Wilk, M.B. An analysis of variance test for normality (complete samples). *Biometrika* **1965**, *52*, 591–611. <https://doi.org/10.1093/biomet/52.3-4.591>. 1042
62. Levene, H. Robust tests for equality of variances. In *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*; Olkin, I., Ed.; Stanford University Press, 1960; pp. 278–292. 1043
63. Cochran, W.G. The comparison of percentages in matched samples. *Biometrika* **1950**, *37*, 256–266. <https://doi.org/10.2307/2332343>. 1044
64. McNemar, Q. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* **1947**, *12*, 153–157. <https://doi.org/10.1007/BF02295996>. 1045
65. Bates, D.; Mächler, M.; Bolker, B.; Walker, S. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* **2015**, *67*, 1–48. <https://doi.org/10.18637/jss.v067.i01>. 1046
66. Benjamini, Y.; Hochberg, Y. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B* **1995**, *57*, 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>. 1047
67. Cohen, J. *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed.; Lawrence Erlbaum Associates, 1988. 1048
68. Lakens, D. Calculating and reporting effect sizes to facilitate cumulative science. *Frontiers in Psychology* **2013**, *4*, 863. <https://doi.org/10.3389/fpsyg.2013.00863>. 1049
69. Rose, D.H.; Meyer, A. *Teaching every student in the digital age: Universal design for learning*; Association for Supervision and Curriculum Development, 2002. 1050
70. Wobbrock, J.O.; Kane, S.K.; Gajos, K.Z.; Harada, S.; Froehlich, J. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing* **2011**, *3*, 1–27. <https://doi.org/10.1145/1952383.1952384>. 1051
71. Treisman, A.M.; Gelade, G. A feature-integration theory of attention. *Cognitive Psychology* **1980**, *12*, 97–136. [https://doi.org/10.1016/0010-0285\(80\)90005-5](https://doi.org/10.1016/0010-0285(80)90005-5). 1052
72. Hullman, J.; Qiao, X.; Correll, M.; Kale, A.; Kay, M. In pursuit of error: A survey of uncertainty visualization evaluation. *IEEE Transactions on Visualization and Computer Graphics* **2019**, *25*, 903–913. <https://doi.org/10.1109/TVCG.2018.2864885>. 1053
73. W3C. *Web Content Accessibility Guidelines (WCAG) 2.2*, 2023. 1054

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